the gross mass according to paragraph 2.1 of this regulation shall be verified by the shipper, either by:
.1 wei
ghing the packed container using calibrated and certified equipment; or
.2 wei
ghing all packages and cargo items, including the mass of pallets, dunnage and other securing material to be packed in the container and adding the tare mass of the container to the sum of the single masses, using a certified method approved by the competent authority of the State in which packing of the container was completed.
5 The s
hipper of a container shall ensure the verified gross mass is stated in the shipping document. The shipping document shall be:
.1 sig
ned by a person duly authorized by the shipper; and
.2 sub
mitted to the master or their representative and to the terminal representative sufficiently in advance, as required by the master or their representative, to be used in the preparation of the ship stowage plan.

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9
RESOLUTION MSC.539(107) (adopted on 8 June 2023)

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6 If t
he shipping document, with regard to a packed container, does not provide the verified gross mass and the master or their representative and the terminal representative have not obtained the verified gross mass of the packed container, it shall not be loaded on to the ship.
Regu
lation 3
Oxygen analysis and gas detection equipment
1 Whe
n transporting a solid bulk cargo which is liable to emit a toxic or flammable gas, or cause oxygen depletion in the cargo space, an appropriate instrument for measuring the concentration of gas or oxygen in the air shall be provided together with detailed instructions for its use. Such an instrument shall be to the satisfaction of the Administration.
2 The A
dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.
Regu
lation 4

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the Administration.
2 The A
dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.
Regu
lation 4

The use of pesticides in ships
App
ropriate precautions shall be taken in the use of pesticides in ships, in particular for the purposes of fumigation.
Regu
lation 5
Stowage and securing
1 Car
go, cargo units and cargo transport units carried on or under deck shall be so loaded, stowed and secured as to prevent as far as is practicable, throughout the voyage, damage or hazard to the ship and the persons on board, and loss of cargo overboard.
2 Car
go, cargo units and cargo transport units shall be so packed and secured within the unit as to prevent, throughout the voyage, damage or hazard to the ship and the persons on board.
3 App
ropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with
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3 App
ropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with
abnormal physical dimensions to ensure that no structural damage to the ship occurs and to maintain adequate stability throughout the voyage.
4 App
ropriate precautions shall be taken during loading and transport of cargo units and cargo transport units on board ro-ro ships, especially with regard to the securing arrangements on board such ships and on the cargo units and cargo transport units and with regard to the strength of the securing points and lashings.
5 Fre
ight containers shall not be loaded to more than the maximum gross weight indicated on the Safety Approval Plate under the International Convention for Safe Containers (CSC), as amended.
6 All c
argoes, other than solid and liquid bulk cargoes, cargo units and cargo transport units, shall be loaded, stowed and secured throughout the voyage in accordance with the Cargo Securing Manual approved by the
10

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stowed and secured throughout the voyage in accordance with the Cargo Securing Manual approved by the
10

Non-c
ommercial use only
Adm
inistration. In ships with ro-ro spaces, as defined in regulation II-2/3.41, all securing of such cargoes, cargo units, and cargo transport units, in accordance with the Cargo Securing Manual, shall be completed before the ship leaves the berth. The Cargo Securing Manual shall be drawn up to a standard at least equivalent to relevant guidelines developed by the Organization.
Part
B

Spec
ial provisions for solid bulk cargoes
Regu
lation 6
Acceptability for shipment
1 Pri
or to loading a solid bulk cargo, the master shall be in possession of comprehensive information on the ships stability and on the distribution of cargo for the standard loading conditions. The method of providing such
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ships stability and on the distribution of cargo for the standard loading conditions. The method of providing such information shall be to the satisfaction of the Administration.
Regu
lation 7
Loading, unloading and stowage of solid bulk cargoes
1 For t
he purpose of this regulation, terminal representative means a person appointed by the terminal or other facility, where the ship is loading or unloading, who has responsibility for operations conducted by that terminal or facility with regard to the particular ship.
To enable the master to prevent excessive stresses in the ships structure, the ship shall be provided with a booklet, which shall be written in a language with which the ships officers responsible for cargo operations are familiar. If this language is not English, the ship shall be provided with a booklet written also in the English language. The booklet shall, as a minimum, include:
.1 sta
bility data, as required by regulation II-1/5-1;
.2 bal

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booklet shall, as a minimum, include:
.1 sta
bility data, as required by regulation II-1/5-1;
.2 bal
lasting and deballasting rates and capacities;
.3 max
imum allowable load per unit surface area of the tank top plating;
.4 max
imum allowable load per hold;
.5 gen
eral loading and unloading instructions with regard to the strength of the ships structure including any limitations on the most adverse operating conditions during loading, unloading, ballasting opera - tions and the voyage;
.6 any s
pecial restrictions such as limitations on the most adverse operating conditions imposed by the Administration or organization recognized by it, if applicable; and
.7 whe
re strength calculations are required, maximum permissible forces and moments on the ships hull during loading, unloading and the voyage.
3 Bef

ore a solid bulk cargo is loaded or unloaded, the master and the terminal representative shall agree on a
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3 Bef
ore a solid bulk cargo is loaded or unloaded, the master and the terminal representative shall agree on a
plan which shall ensure that the permissible forces and moments on the ship are not exceeded during loading or unloading, and shall include the sequence, quantity and rate of loading or unloading, taking into consideration the speed of loading or unloading, the number of pours and the deballasting or ballasting capability of the ship. The plan and any subsequent amendments thereto shall be lodged with the appropriate authority of the port State.

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4 The m
aster and terminal representative shall ensure that loading and unloading operations are conducted in accordance with the agreed plan.
5 If du
ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are
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accordance with the agreed plan.
5 If du
ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are
likely to become so if the loading or unloading continues, the master has the right to suspend operation and the obligation to notify accordingly the appropriate authority of the port State with which the plan has been lodged. The master and the terminal representative shall ensure that corrective action is taken. When unloading cargo, the master and terminal representative shall ensure that the unloading method does not damage the ships structure.
6 The m
aster shall ensure that ships personnel continuously monitor cargo operations. Where possible, the ships draught shall be checked regularly during loading or unloading to confirm the tonnage figures supplied. Each draught and tonnage observation shall be recorded in a cargo logbook. If significant deviations from the
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